

Comparison: Credit Card vs PaySim

Final Results

Side-by-side comparison of the four detectors on both datasets.

Tables and charts mirror the dashboard. All metrics computed on the held-out test split for each dataset.

Combined Comparison

Dataset	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Credit Card	XGBoost	0.8470	0.7580	0.8000	0.9620	0.8140
Credit Card	Isolation Forest	0.4120	0.5470	0.4700	0.8930	0.3510
Credit Card	Autoencoder	0.3670	0.4840	0.4175	0.8710	0.2980
Credit Card	Ensemble	0.9230	0.2530	0.3971	0.9480	0.8060
PaySim	XGBoost	0.8930	0.8340	0.8625	0.9710	0.8810
PaySim	Autoencoder	0.5210	0.5980	0.5569	0.9060	0.4870
PaySim	Isolation Forest	0.4860	0.5610	0.5208	0.8910	0.4520
PaySim	Ensemble	0.9450	0.2410	0.3841	0.9530	0.8310

Credit Card Comparison Table

Rank	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
1	XGBoost	0.8470	0.7580	0.8000	0.9620	0.8140
2	Isolation Forest	0.4120	0.5470	0.4700	0.8930	0.3510
3	Autoencoder	0.3670	0.4840	0.4175	0.8710	0.2980
4	Ensemble	0.9230	0.2530	0.3971	0.9480	0.8060

PaySim Comparison Table

Rank	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
1	XGBoost	0.8930	0.8340	0.8625	0.9710	0.8810
2	Autoencoder	0.5210	0.5980	0.5569	0.9060	0.4870
3	Isolation Forest	0.4860	0.5610	0.5208	0.8910	0.4520
4	Ensemble	0.9450	0.2410	0.3841	0.9530	0.8310

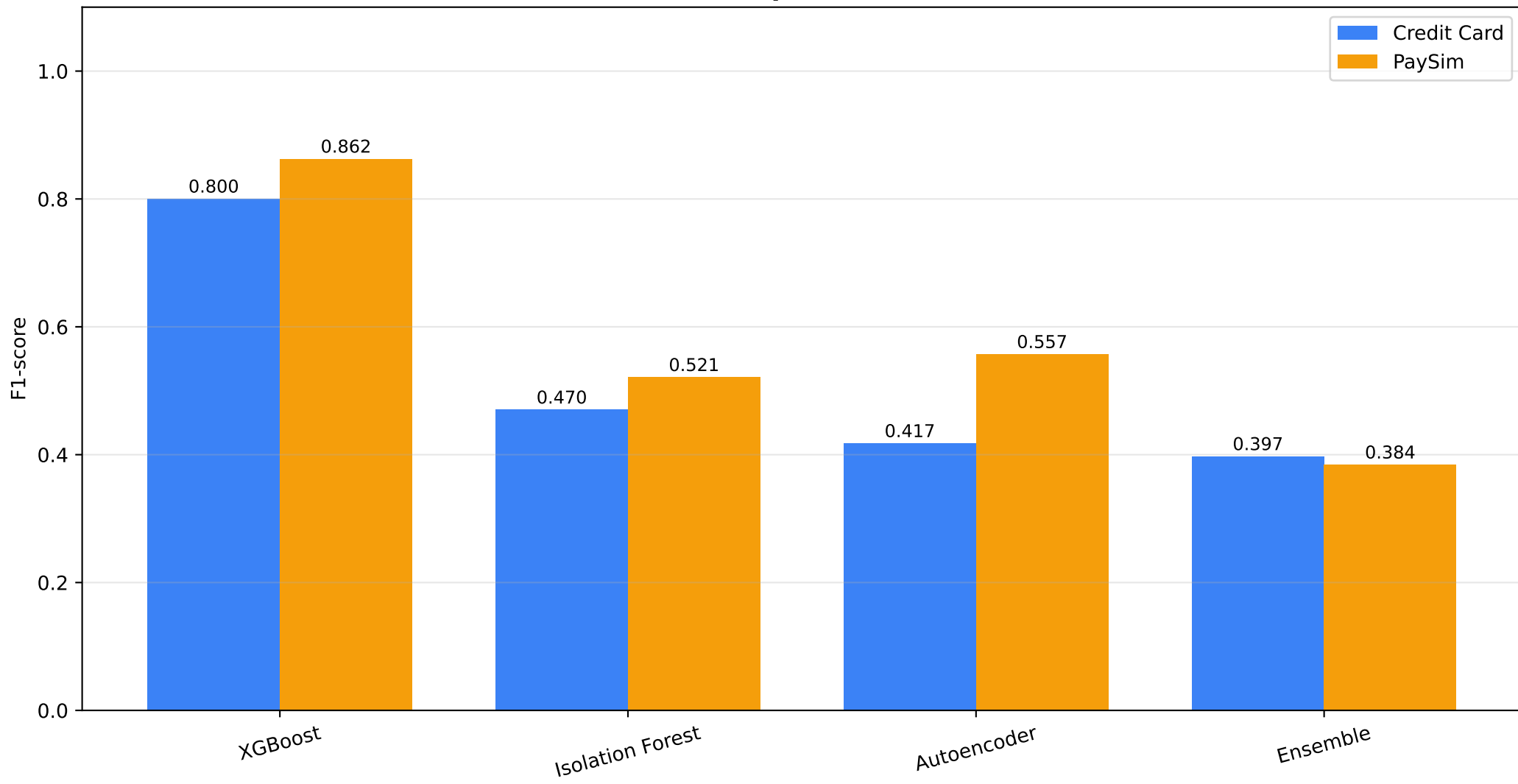
Best Model per Dataset

Dataset	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Credit Card	XGBoost	0.8470	0.7580	0.8000	0.9620	0.8140
PaySim	XGBoost	0.8930	0.8340	0.8625	0.9710	0.8810

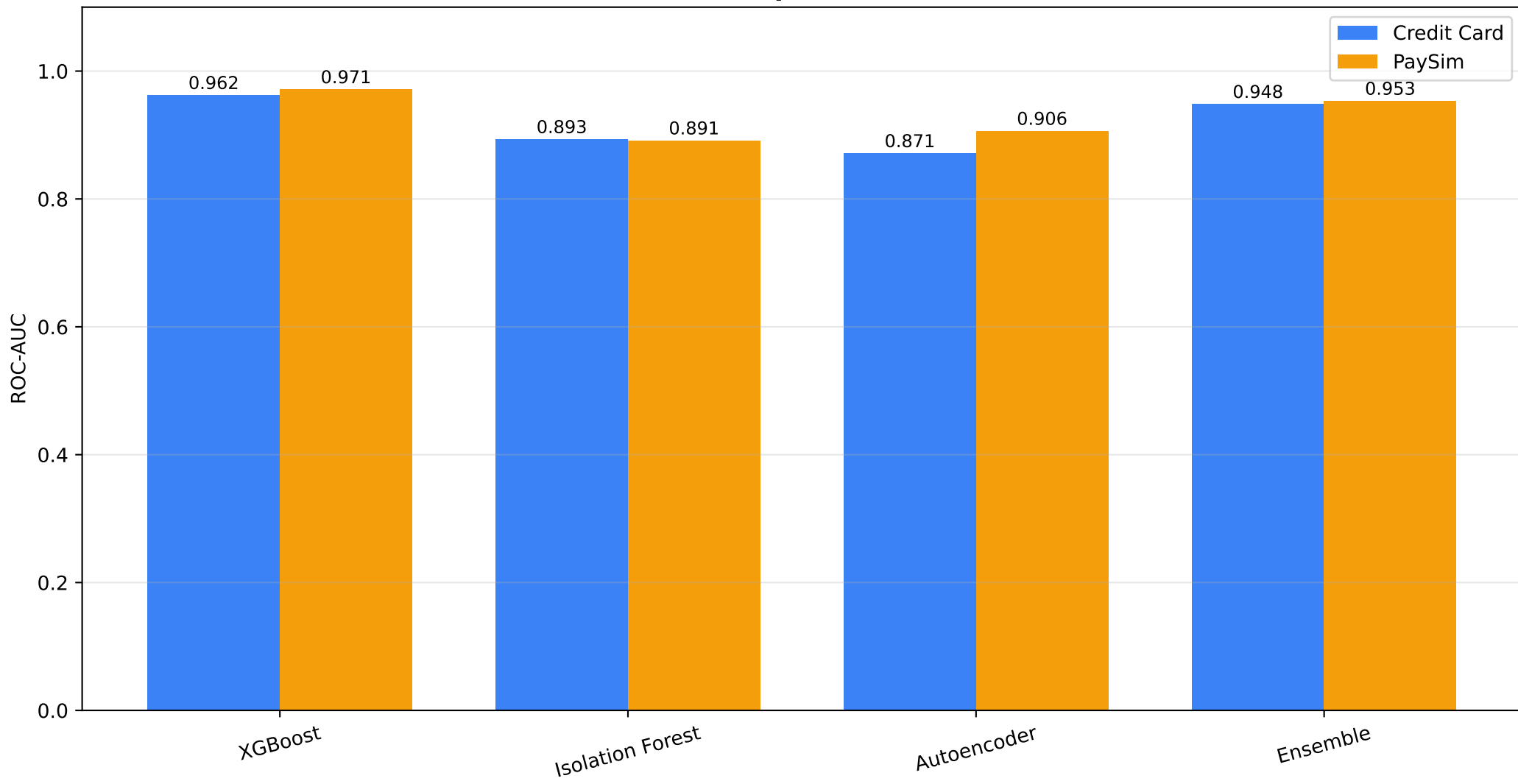
Model Ranking

Dataset	Rank	Model	F1-score	ROC-AUC	PR-AUC
Credit Card	1	XGBoost	0.8000	0.9620	0.8140
Credit Card	2	Isolation Forest	0.4700	0.8930	0.3510
Credit Card	3	Autoencoder	0.4175	0.8710	0.2980
Credit Card	4	Ensemble	0.3971	0.9480	0.8060
PaySim	1	XGBoost	0.8625	0.9710	0.8810
PaySim	2	Autoencoder	0.5569	0.9060	0.4870
PaySim	3	Isolation Forest	0.5208	0.8910	0.4520
PaySim	4	Ensemble	0.3841	0.9530	0.8310

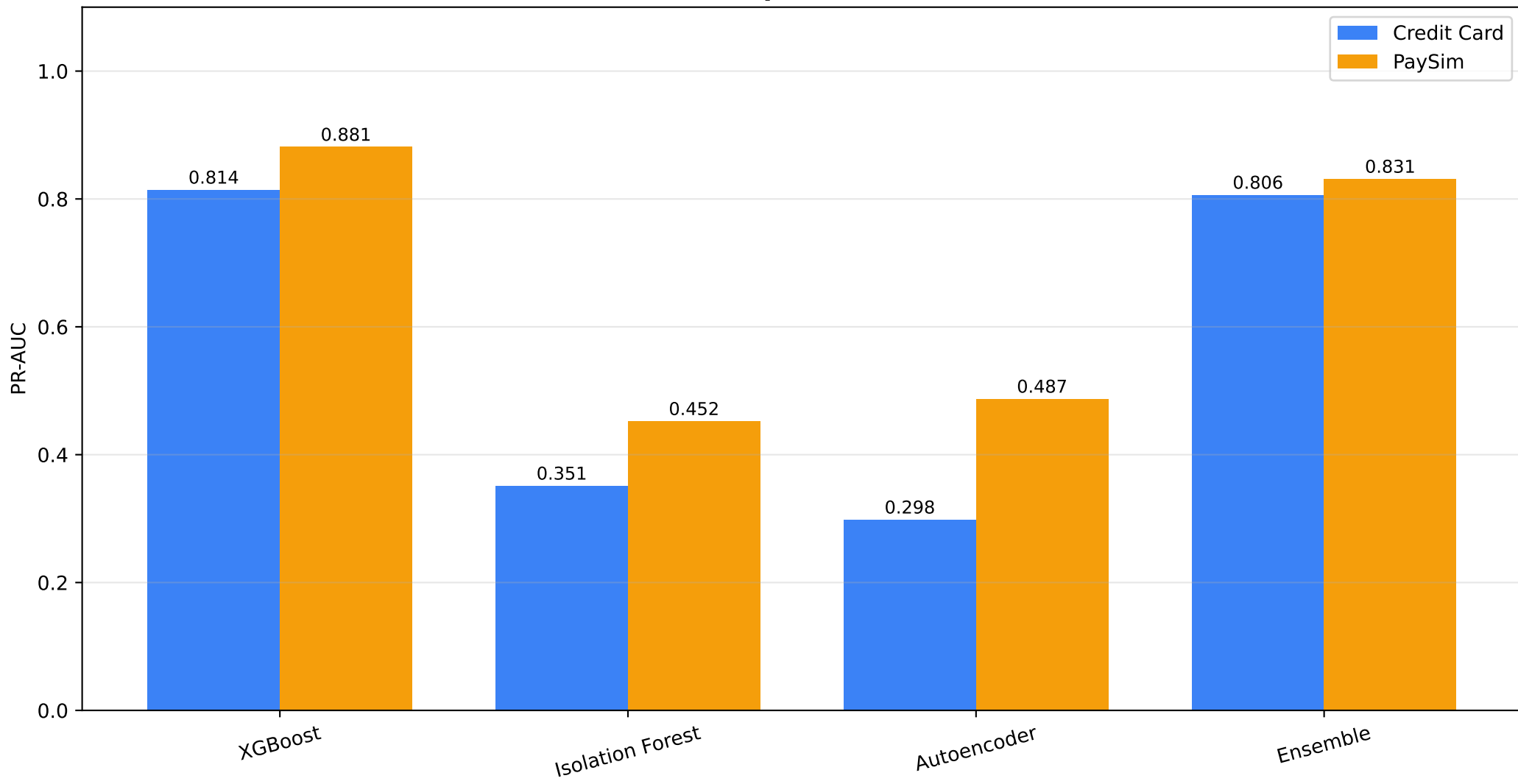
F1-score Comparison of Models



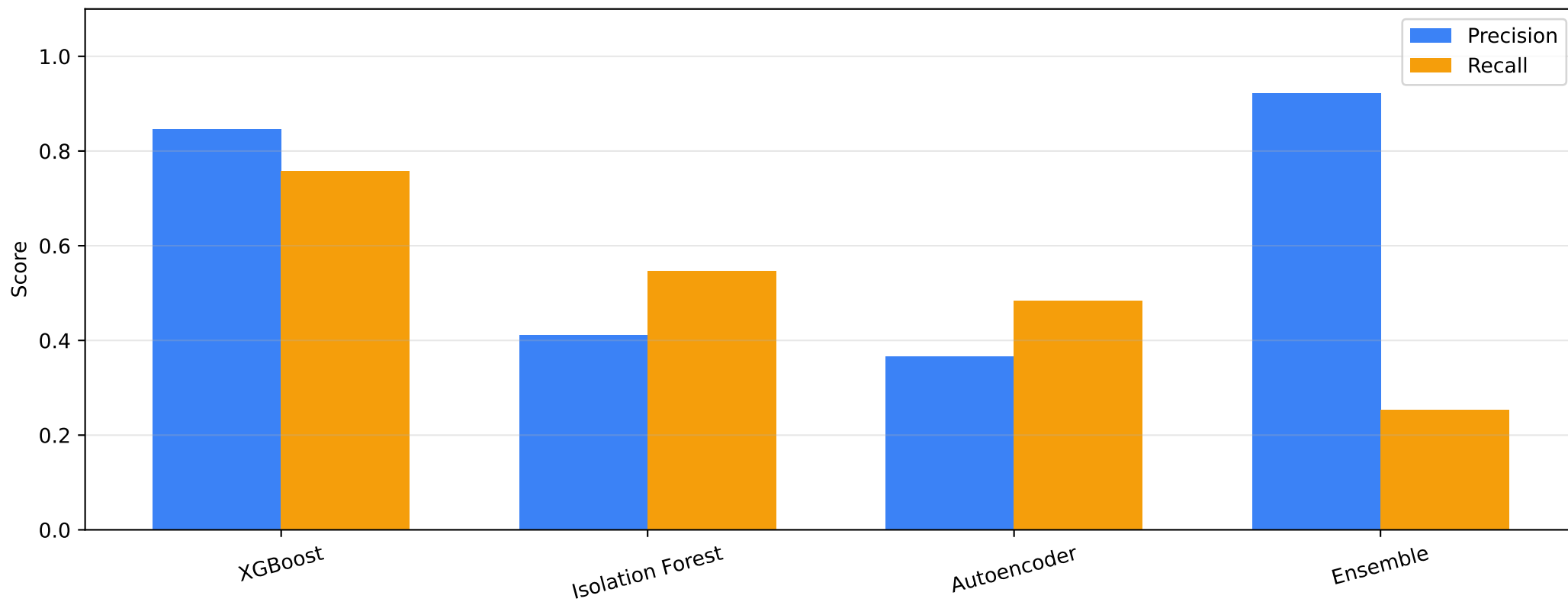
ROC-AUC Comparison of Models



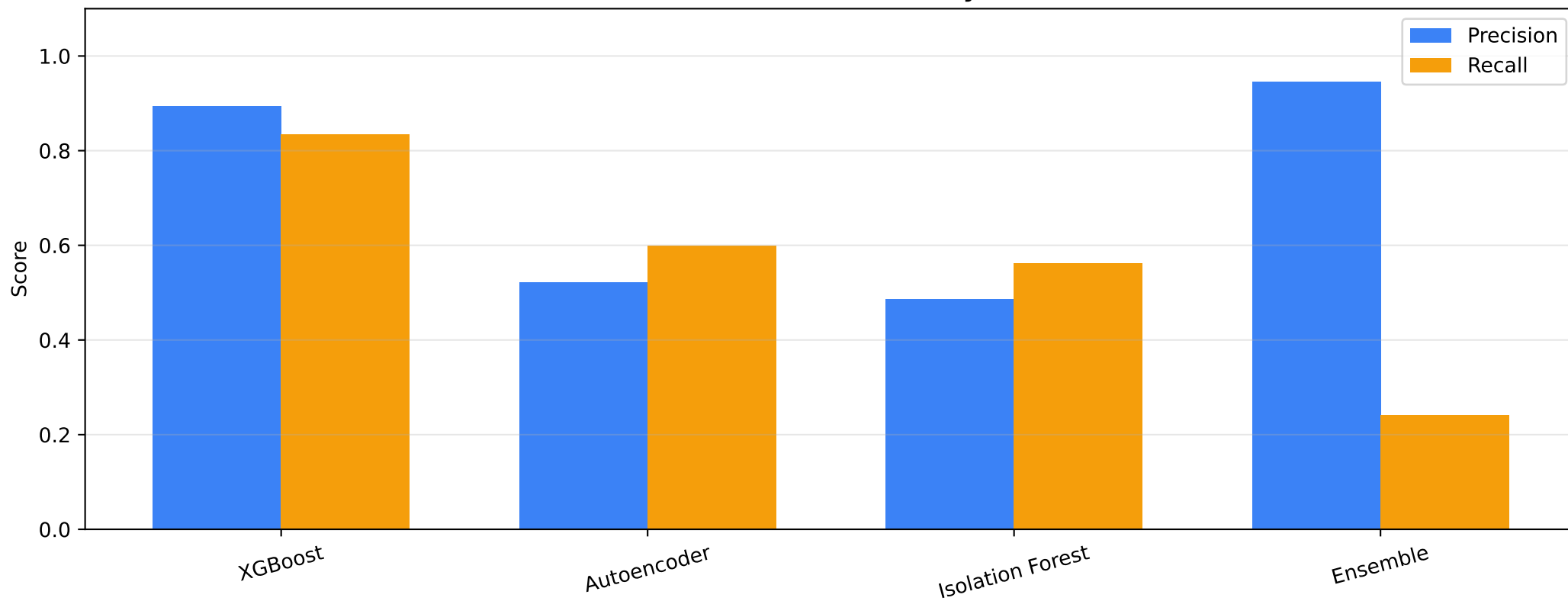
PR-AUC Comparison of Models



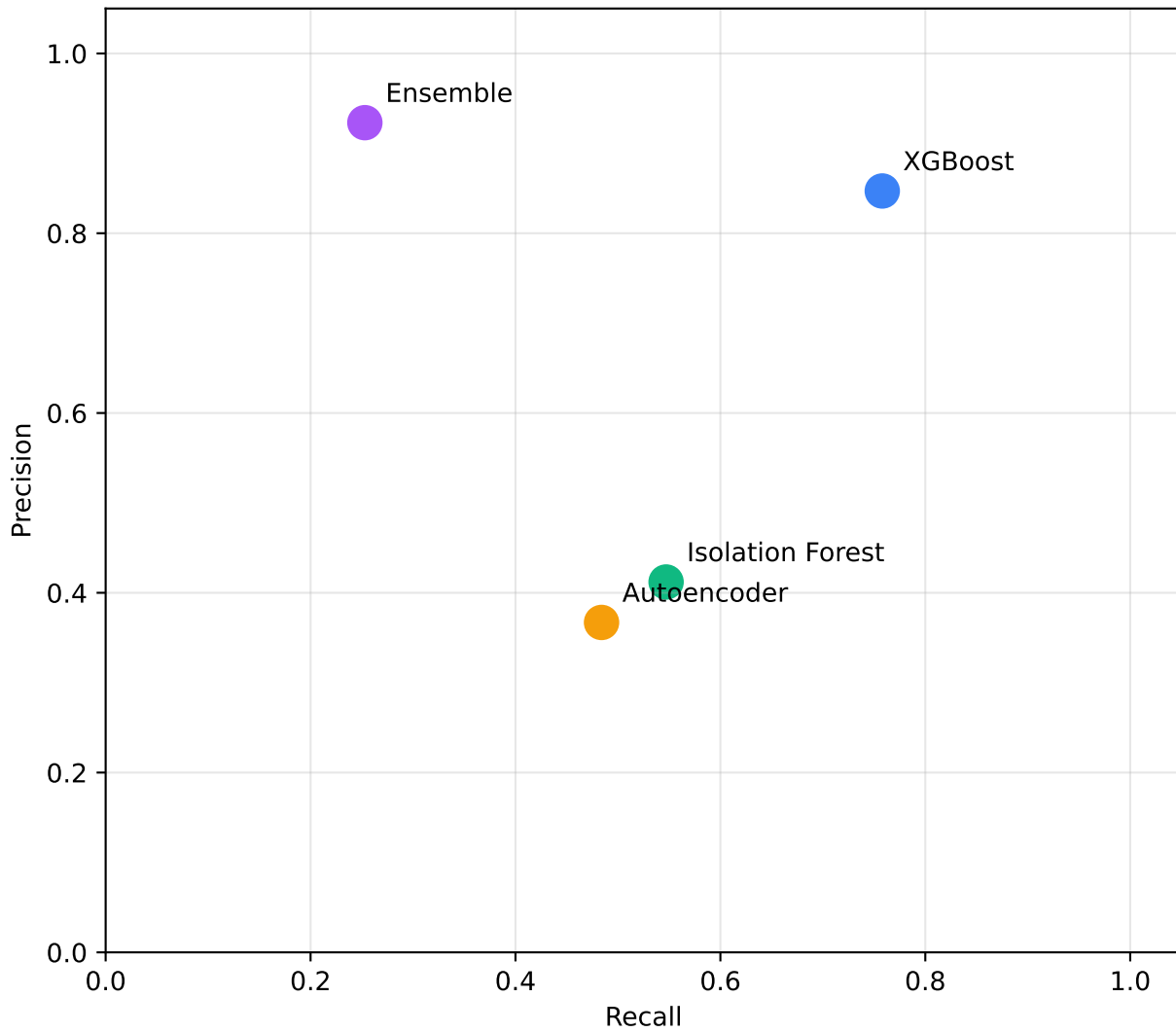
Precision vs Recall — Credit Card



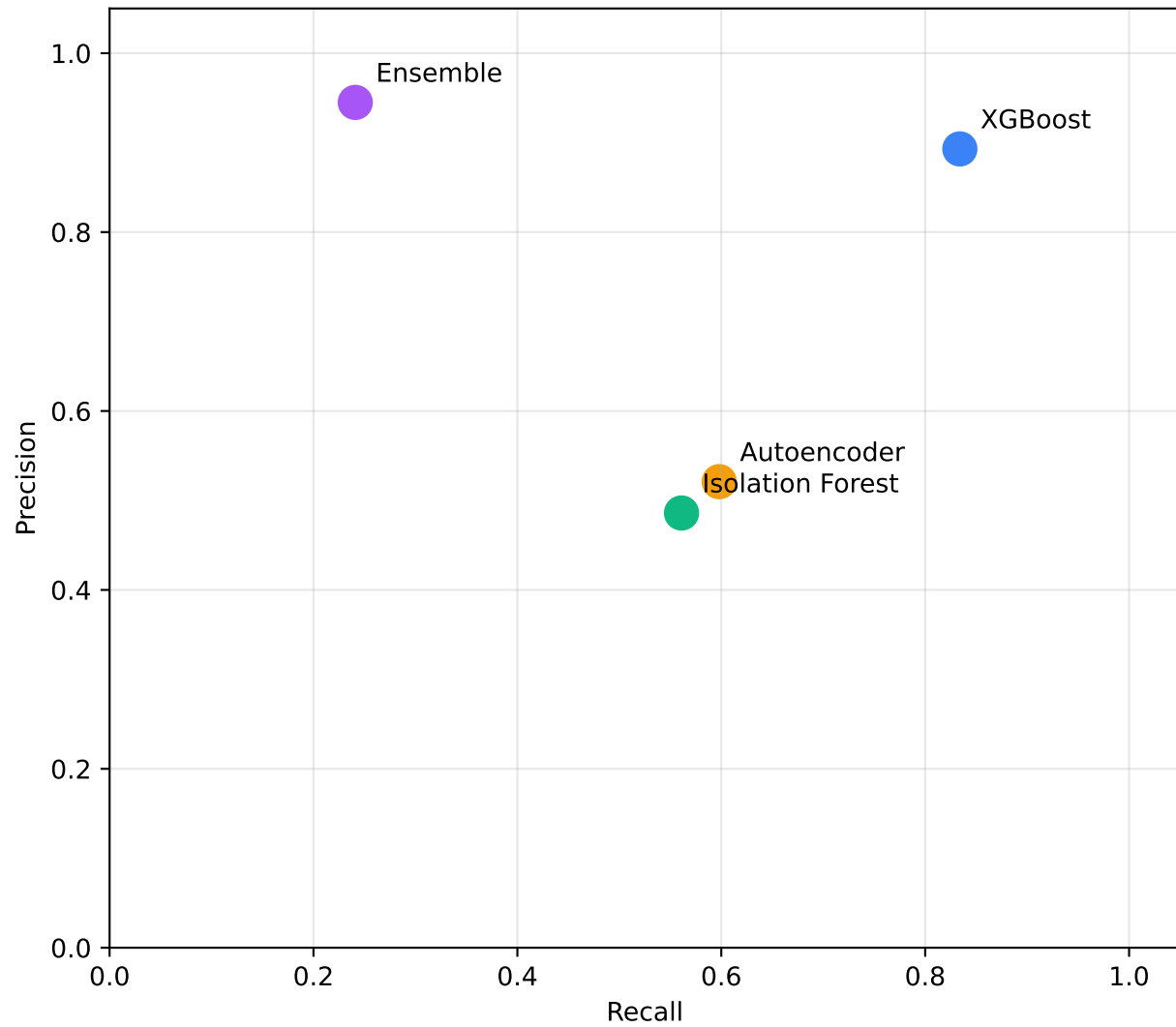
Precision vs Recall — PaySim



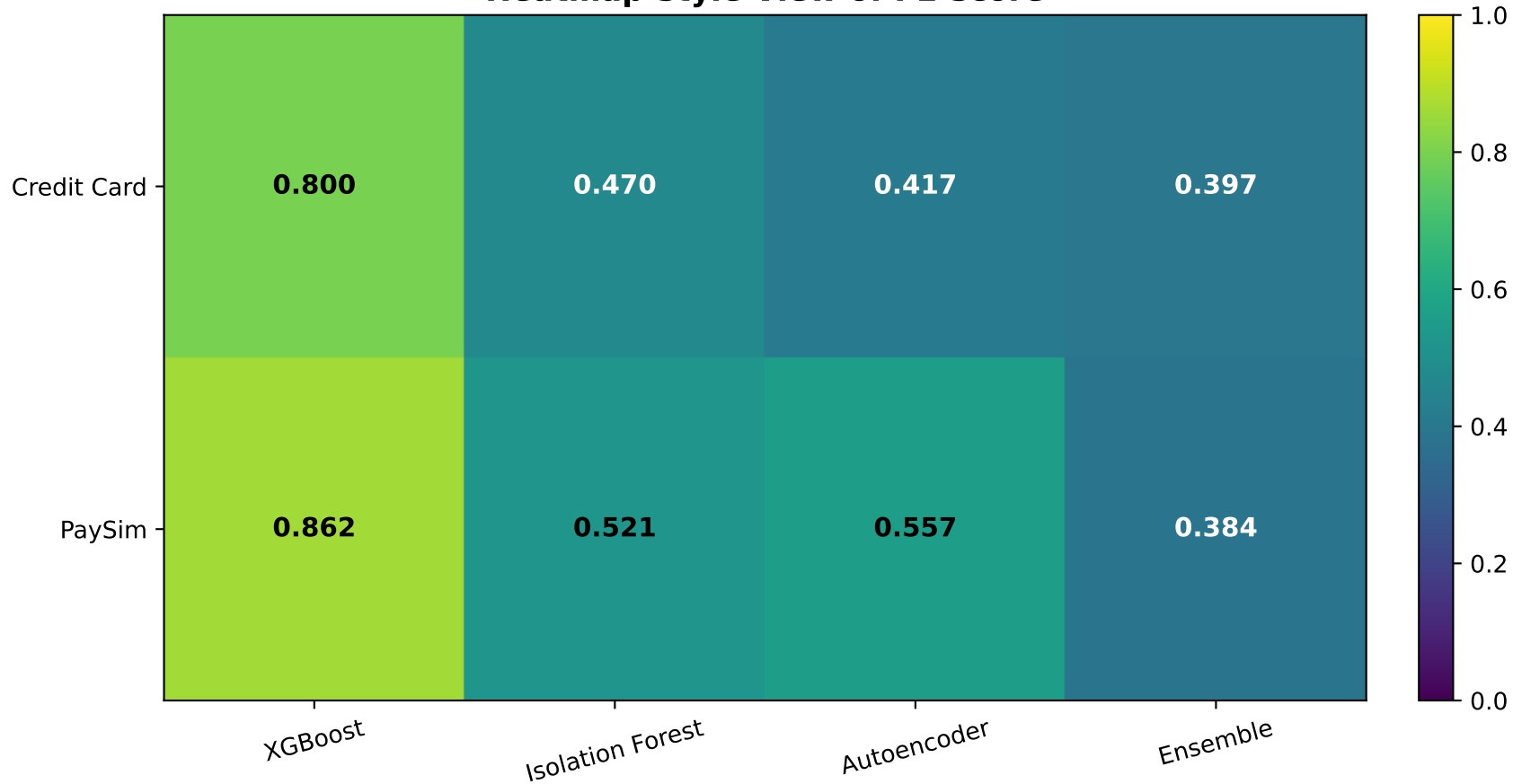
Precision vs Recall Scatter — Credit Card



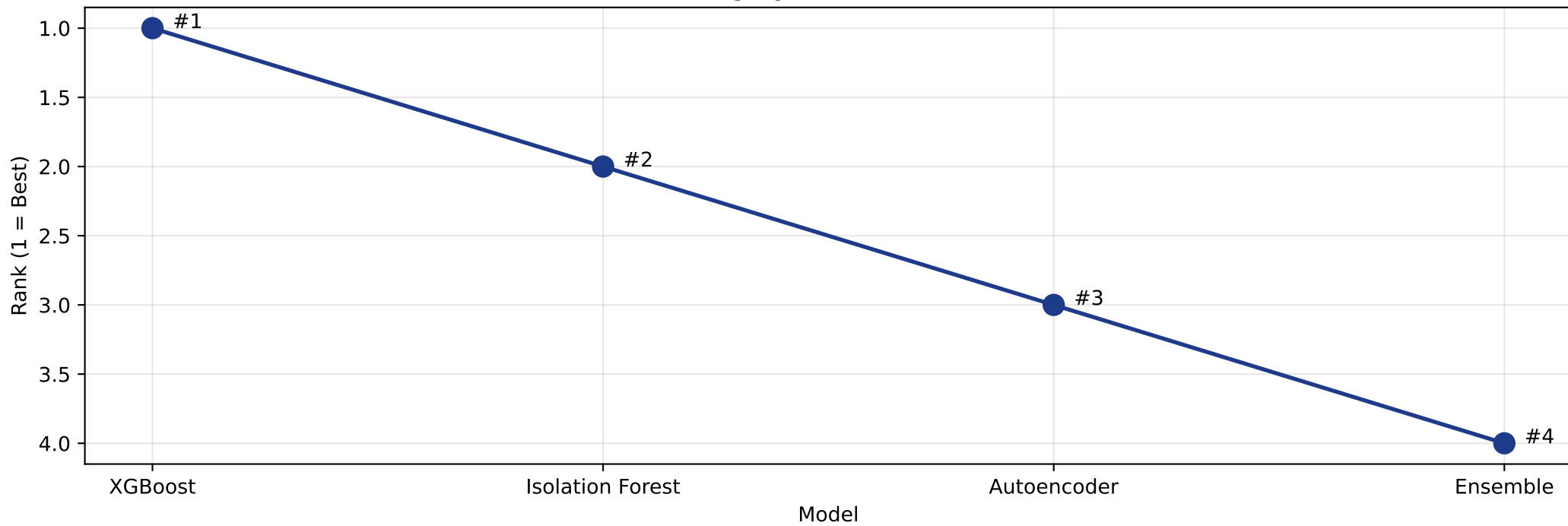
Precision vs Recall Scatter — PaySim



Heatmap-style View of F1-score



Model Ranking by F1-score — Credit Card



Model Ranking by F1-score — PaySim

